

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

103001058 - Generative Ai And Language Models

DEGREE PROGRAMME

10BA - Master Universitario En Ciencia De Datos

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

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1. Description

1.1. Subject details

Name of the subject	103001058 - Generative Ai And Language Models
No of credits	4.5 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	10BA - Master Universitario en Ciencia de Datos
Centre	10 - E.T.S. De Ingenieros Informáticos
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Emilio Serrano Fernandez (Subject coordinator)	2201	emilio.serrano@upm.es	Sin horario. https://dia.fi.upm.es/personaldia/emilio/err/
Damiano Zanardini	2205	damiano.zanardini@upm.es	Sin horario. https://dia.fi.upm.es/personaldia/damiano/

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Basic knowledge of Python.

4. Skills and learning outcomes *

4.1. Skills to be learned

CB07 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio

CB10 - Que los estudiantes posean las habilidades de aprendizaje que les permitan continuar estudiando de un modo que habrá de ser en gran medida autodirigido o autónomo.

CECD07 - Capacidad para desarrollar de modelos inteligentes basados en datos

CG07 - Aplicación de los últimos o más novedosos métodos para resolver problemas que, posiblemente, involucren a otras disciplinas

CG08 - Capacidad de pensamiento de forma creativa para desarrollar aproximaciones y métodos nuevos y originales

CG09 - Integración del conocimiento de distintos campos de estudio

CG10 - Apreciación de los límites del conocimiento actual y de la aplicación práctica de la última tecnología

CG11 - Conocimiento y comprensión de la informática para crear modelos, así como sistemas y procesos de información complejos

4.2. Learning outcomes

RA46 - Design and evaluate effective prompts for generative AI tasks, including few-shot, zero-shot, and chain-of-thought prompting

RA45 - Understand the principles, architectures, and limitations of generative models, including large language models

RA33 - understand and apply information retrieval systems

RA47 - Propose innovative uses of generative AI for problem-solving and creative expression across domains

RA35 - Identify areas of application where deep learning techniques can be used

RA36 - Apply machine learning software tools for practical problems related to deep learning

RA32 - Understand and design information extraction systems

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

This course provides a deep and practical introduction to one of the most transformative areas in modern artificial intelligence: Generative AI and Language Models. Students will explore how machines can generate text, images, and other forms of media by leveraging powerful deep learning architectures.

Generative AI refers to AI systems capable of producing novel content based on human input - commonly called prompts. Language models, particularly large-scale models like the GPT series lie at the heart of these systems. They not only process and generate human language but can also serve as a foundation for multimodal applications that integrate text, images, and audio.

These technologies are evolving rapidly and becoming increasingly accessible through websites, APIs, and open-source tools. As a result, their influence is spreading across many sectors. They are transforming creative industries, enhancing education, and driving innovation in fields such as scientific research, healthcare, and more.

The course explores both the theoretical foundations and practical techniques needed to build generative AI solutions.

5.2. Syllabus

1. Introduction to Generative AI and Language Models
2. Distributional Semantics and Word Embeddings
3. Sequential Models: RNNs, LSTMs, GRUs
4. Sequence-to-Sequence Models and the Transformer Architecture
5. Pretrained (Large) Language Models: GPT, BERT, T5, and beyond
6. Conversational AIs: ChatGPT, Gemini, and beyond
7. Frameworks for Building Generative AI Applications
8. Prompt Engineering: Few-shot, Chain-of-Thought, and beyond
9. Advanced Architectures: Retrieval-Augmented Generation (RAG) and Agentic Design Patterns
10. Multimodal Generative AI: text-to-image, image-to-text, and beyond

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Topic 1 Duration: 02:00 Lecture			
2	Topic 2 Duration: 02:00 Lecture			
3	Topic 3 Duration: 02:00 Lecture			
4	Topic 4 Duration: 02:00 Lecture			
5	Topic 5 Duration: 02:00 Lecture			
6	Topic 6 Duration: 02:00 Lecture			
7	Topic 7 Duration: 02:00 Lecture			
8	Topic 8 Duration: 02:00 Lecture			
9	Topic 9 Duration: 02:00 Lecture			
10	Topic 10 Duration: 02:00 Lecture			
11	Theoretical and Practical Exam topics 1-10 Duration: 02:00 Additional activities			Theoretical and Practical Exam topics 1-10 Written test Progressive assessment Presential Duration: 02:00
12	Practical Project Development Duration: 02:00 Laboratory assignments			
13	Practical Project Development Duration: 02:00 Laboratory assignments			

14	Practical Project Development Duration: 02:00 Laboratory assignments			
15	Presentations of Practical Projects Duration: 02:00 Additional activities			Presentations of Practical Projects Group work Progressive assessment Presential Duration: 02:00
16				
17				Theoretical and Practical Exam topics 1-10 Written test Global examination Presential Duration: 02:00 Practical Project Individual work Global examination Presential Duration: 02:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
11	Theoretical and Practical Exam topics 1-10	Written test	Face-to-face	02:00	70%	5 / 10	CB07 CB10 CG07 CG08 CG09 CG10 CG11 CECD07
15	Presentations of Practical Projects	Group work	Face-to-face	02:00	30%	5 / 10	CB07 CB10 CG07 CG08 CG09 CG10 CG11 CECD07

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Theoretical and Practical Exam topics 1-10	Written test	Face-to-face	02:00	70%	5 / 10	CB07 CB10 CG07 CG08 CG09 CG10 CG11 CECD07
17	Practical Project	Individual work	Face-to-face	02:00	30%	5 / 10	CB07 CB10 CG07 CG08 CG09 CG10 CG11 CECD07

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Theoretical and Practical Exam topics 1-10	Written test	Face-to-face	02:00	70%	5 / 10	CB07 CB10 CG07 CG08 CG09 CG10 CG11 CECD07
Practical Project	Individual work	Face-to-face	02:00	30%	5 / 10	CB07 CB10 CG07 CG08 CG09 CG10 CG11 CECD07

7.2. Assessment criteria

- Both classroom activities and assessment methods are designed for groups of up to 50 students per classroom. If enrollment exceeds this limit, the practical project component may be excluded, and the syllabus may be adjusted to make use of the available time. - Qualifications of 5 or higher in partial exams and practical projects are kept for the global and extraordinary evaluation.
- You can only opt for a second or third evaluation of partial exams and practical projects if they have received grades lower than 5 or have not been submitted.
- If the minimum passing grade is not achieved in one of the evaluated parts, the grade for the entire course will be the grade of the part that was not passed.
- The instructions for practical projects, as well as the type of written exam (multiple choice, short answer, essay) may vary in different assessments (progressive, global, and extraordinary).
- In the event that a written exam is evaluated through quizzes on the Moodle platform with random questions from

a question bank, the solutions to the questions will not be published since it is not allowed by this type of exam. Besides, in this exam modality and to enhance anti-cheating measures, access to the quiz will be sequential (only one question can be viewed per screen, and it will not be possible to go back once a question is answered).

- The rights and duties of university students are developed in the Statutes of the Polytechnic University of Madrid (BOCM of November 15, 2010) and in the University Student Statute (RD 1791/2010 of December 30). Article 124 (a) of the statutes of the UPM establishes as the student's duty..."to follow responsibly and profitably the process of training, knowledge acquisition, and learning corresponding to their condition as a university student"... and article 13 of the Statute of the University Student, in point (d) it also specifies as the duty of the university student "to refrain from the use or cooperation in fraudulent procedures in the evaluation tests, in the work that is carried out, or in official documents of the university". In the event that in the development of the evaluation tests the breach of the duties as a university student is appreciated, the coordinator of the subject may notify the Director if the Center, who in accordance with the provisions of article 74 (n) of the UPM Statutes has powers to "Propose the initiation of disciplinary proceedings against any member of the School or Faculty".

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
UPM Moodle	Web resource	
[Zhao, W. X., et al., 2023]	Bibliography	Zhao, W. X., et al. (2023). A Survey of Large Language Models. arXiv. https://arxiv.org/abs/2303.18223
[Ng 2023]	Web resource	Online course. Generative AI for Everyone, https://www.deeplearning.ai/courses/generative-ai-for-everyone/
Bibliografía	Bibliography	Selected bibliography (articles and textbooks).

9. Other information

9.1. Other information about the subject

The course uses the Moodle platform to provide students with information and documentation, as well as to announce evaluation schedules and publish grades.

This course contributes to the advancement of several Sustainable Development Goals (SDGs) defined by the United Nations 2030 Agenda, in particular:

- SDG 4 - Quality Education: Through the study and application of Deep Learning technologies in Natural Language Processing (NLP), students acquire advanced AI skills that can be applied to improve access to information, reduce language barriers, and support personalized education (e.g., intelligent tutoring systems or automated educational content generation).
- SDG 9 - Industry, Innovation and Infrastructure: The knowledge gained on advanced models such as Transformers, LLMs, and GNNs equips students to develop innovative solutions in strategic sectors that rely on natural language processing, including public services, healthcare, transportation, and applied language technologies.
- SDG 10 - Reduced Inequalities: By exploring tools like machine translation, multilingual conversational systems, and semantic access to information, the course promotes technologies that can help reduce inequalities in access to knowledge and services, especially for populations facing linguistic or technological barriers.
- SDG 16 - Peace, Justice and Strong Institutions: The use of NLP techniques for understanding legal texts, promoting institutional transparency, or moderating content can contribute to strengthening democratic systems and ensuring equitable access to information.

The course follows an active and hands-on methodology, combining theoretical sessions with workshops and applied projects. Current tools such as Hugging Face and ChatGPT are used. Moreover, it emphasizes the ethical and responsible use of artificial intelligence, encouraging reflection on issues like bias and fairness in language processing systems.